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Optimizing Underbalance Coiled Tubing Descaling Operations: Leveraging Downhole Telemetry for Operational Efficiency

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Abstract

Scale accumulation in oil and gas wells presents significant challenges, particularly in low bottom hole pressure environments where conventional overbalanced cleanout techniques fail, risking fluid losses that compromise formation integrity and increase intervention costs. This paper demonstrates how underbalance descaling, enhanced with real-time downhole telemetry, mitigates these risks by minimizing fluid losses, protecting the reservoir, and reducing chemical consumption, water usage, and carbon emissions. Case studies illustrate how this approach reduces well intervention costs while maintaining high operational efficiency.

Descaling operations were performed using Coiled Tubing (CT) with a bottom-hole assembly (BHA) tailored for sour well conditions with severe iron sulfide scale deposits. The BHA incorporated real-time telemetry, enabling continuous monitoring of pressure, temperature, torque, tension-compression, and vibration. This data-driven approach enabled:

1. Improved efficiency: real-time insights optimized fluid selection and rate of penetration by monitoring key parameters like tension and vibration.
2. Risk mitigation: downhole pressure monitoring minimized the risk of stuck CT pipes.
3. Reservoir protection: underbalanced conditions prevented removed scale from re-entering the formation, safeguarding integrity.

Overall, this methodology significantly improved operational performance, reduced risks, and ensured reservoir integrity and improved environmental and regulatory compliance.

The optimization strategy delivered impressive results across six wells (Well-A through Well-F):

- Fluid consumption: total fluid volume reduced by 20% to 70%, including up to 70% reduction in water usage.
- Penetration rate and intervention time: penetration rates improved by 200%, reducing intervention time by 66%.
- Environmental impact: carbon emissions decreased by up to 66%.

- Cost and safety: lower chemical waste and reduced costs, along with enhanced safety through real-time monitoring. These outcomes highlight the transformative role of telemetry in optimizing descaling operations while reducing environmental and operational risks.

This paper introduces a groundbreaking approach to underbalance descaling using real-time telemetry. It addresses challenges in sour wells and low-pressure environments, providing scalable solutions for operational optimization, cost reduction, and environmental sustainability. This approach sets a new standard for efficient and environmentally conscious scale management.

Introduction

In our previous work (Ayesh et al. 2025), titled "The First Worldwide Dynamic Solution for Iron Sulfide Removal Using Coiled Tubing", we introduced an innovative underbalanced descaling system designed to tackle the significant challenges posed by iron sulfide scale accumulation in hydrocarbon wells. This method emphasized maintaining underbalanced conditions within a closed-loop system, thereby improving operational safety and efficiency while preserving formation integrity.

Building upon this foundation, the present study advances the methodology by incorporating real-time downhole telemetry. This integration enables continuous monitoring of key parameters—pressure, temperature, weight on bit (WOB), vibration, torque, and tension/compression—allowing for timely data-driven decisions during the operation. By leveraging telemetry, the descaling process was optimized in terms of fluid and nitrogen consumption, rate of penetration, intervention time, and carbon emissions.

The effectiveness of this approach is demonstrated through six case studies. These results highlight how telemetry-guided operations can mitigate risks such as stuck pipe, reduce resource usage, and improve environmental performance. This paper sets a new benchmark for efficient and sustainable scale management in sour and low-pressure well environments.

Background

Challenges of Iron Sulfide Scale Accumulation

Iron sulfide scale poses significant operational and safety challenges, particularly in mature, sour wells with high water cut. The presence of H_2S accelerates FeS formation, leading to reduced productivity, stuck pipe risks, and repeated interventions. Its hard, adherent nature makes it difficult to remove using conventional methods, often resulting in extended non-productive time (NPT) and elevated operational costs.

Limitations of Conventional Techniques

Traditional overbalanced techniques often involve introducing fluids into the wellbore under conditions that exceed the formation pressure. This can lead to formation damage, increased risk of well control issues, and higher HSE risks. Overbalanced operations may also result in greater fluid losses and environmental impacts due to the disposal of large volumes of spent fluids. The need for a safer, more efficient approach has driven the development of underbalanced descaling systems.

Overview of Underbalanced Descaling Methodology

The underbalanced descaling technique maintains the wellbore pressure below formation pressure using nitrified fluids in a closed-loop system. This approach minimizes formation damage and improves operational control. In wells with sufficient formation energy, hydrocarbon flow can aid in scale transport, further reducing the need for aggressive pumping. Combined with real-time telemetry, this method allows early detection of circulation loss, scale buildup, and other downhole risks—supporting safer, more efficient, and environmentally responsible scale management.

Methodology

The optimization of underbalance coiled tubing descaling operations relied on a systematic approach to integrating real-time telemetry into the workflow. Initially, early job iterations involved trial-and-error methods as operators gained familiarity with the telemetry data and its significance in guiding the operational decision-making. However, with progressive field applications, operational parameters such as torque, vibration, pressure, and temperature became the basis for fine-tuning the descaling methodology.

- Critical Operational Parameters Monitored

  i. Pressure: Monitoring annular and tubing pressure ensured that operations remained underbalanced, helping prevent fluid invasion and formation damage. Pressure fluctuations also helped detect early signs of scale pack-off or fluid losses.
  ii. Temperature: Downhole temperature provided insight into fluid flow behavior and thermal stability. A gradual temperature drop, especially when paired with surface pressure decline, often indicated circulation losses or scale accumulation risks
  iii. Tension/Compression & Weight on Bit (WOB): These values helped evaluate mechanical interaction between the tool and scale. Abnormally high WOB indicated possible obstruction, while sharp drops signaled sudden breakthrough or free movement
  iv. Vibration: Continuous vibration monitoring served as an early indicator of turbine rotational health. Low vibration levels typically accompanied turbine stalls or tool resistance due to hard scale.
  v. Torque: monitoring torque helps ensure optimal performance of the BHA and indicates the health of BHA during the descaling process while low torque coupled with insignificant WOB can indicate free path below the BHA.

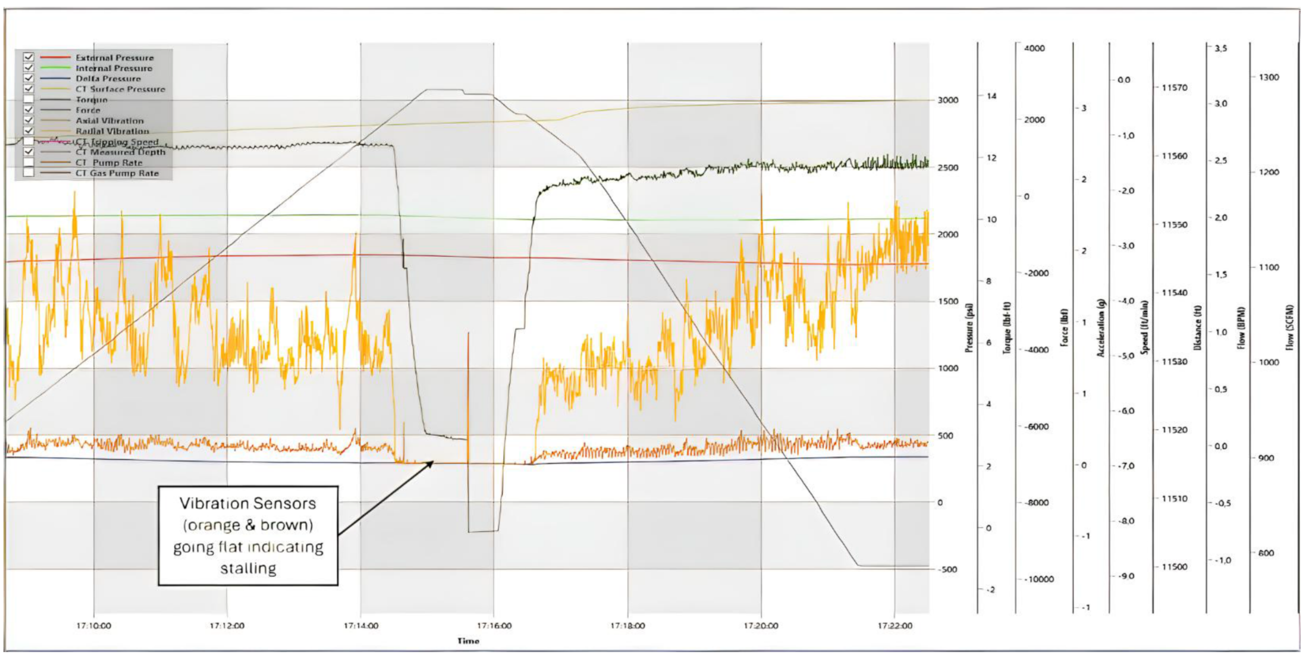
  For readers seeking a deeper understanding of downhole telemetry systems in coiled tubing operations, several SPE papers provide valuable insights into the technologies, measurement parameters, and field applications. Recommended references include SPE-187374-MS, SPE-224070-MS, SPE-174850-MS, and SPE-188421-MS, which collectively cover wireline-based and fiber-optic telemetry systems, real-time data acquisition, and their impact on operational efficiency.

- Real-Time Telemetry and Data Utilization

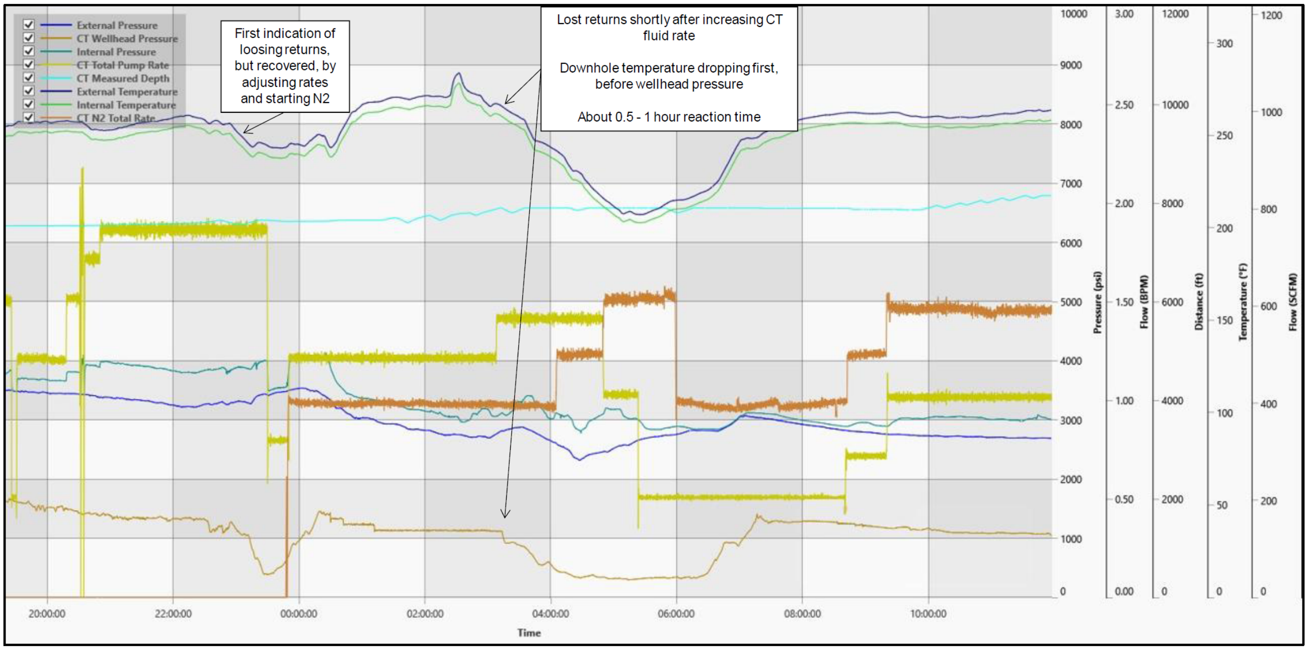
  The BHA was equipped with an integrated telemetry system that transmitted real-time data on pressure, temperature, torque, vibration, and WOB. This enabled the surface team to adapt to dynamic wellbore conditions and improve descaling efficiency

  1. Descaling Confirmation via Torque and Vibration: A rise in torque correlated with active scale cutting. Vibration analysis indicated turbine efficiency; low vibration often meant stalling, prompting flow rate adjustments
  2. WOB and Torque Correlation: The combination of increased WOB and torque was used to determine whether additional passes were required or whether scale removal was complete.
  3. Underbalance Control via Pressure Monitoring: Downhole pressure readings were critical in maintaining underbalanced conditions. Drops in both pressure and temperature were interpreted as fluid loss events, triggering corrective action such as nitrogen injection or rate reduction
  4. Speed Optimization: CT movement speed was adjusted based on torque, vibration & WOB feedback. In clean intervals, speeds were increased to 20–30 fpm to reduce intervention time without risking tool integrity. In high-scale zones, lower speeds preserved tool life and ensured more thorough cleaning.

- Stepwise Optimization Strategy
    1. Baseline Data Collection: Prior to descaling, baseline telemetry readings (pressure, temperature, torque, vibration) were established after tagging scale. These served as reference points to detect active descaling or risks such as loss of circulation.
    2. Real-Time Monitoring and Adaptive Control: Operators continuously analyzed telemetry data. When torque and vibration spiked, penetration was slowed to enhance cleaning efficiency. If parameters stabilized, speed was increased—up to 30 fpm—balancing efficiency and tool wear
    3. Circulation Losses Mitigation: A drop in both pressure and temperature indicated fluid losses into the formation. Nitrogen and base fluid rates were adjusted to maintain underbalance and reduce the risk of scale fallback.
    4. Operational Efficiency and Learning Integration: With each well, telemetry interpretation improved. Operators adapted faster to downhole changes, enhancing decision-making and reducing CT runs, while enabling predictive maintenance and better resource planning



Graph 1—Constant Vibration Reading Indicating Stalling



Graph 2—Temperature and Pressure Drop indicating Loss of Circulation

The integration of real-time telemetry transformed underbalance descaling from a trial-and-error approach into a precise and controlled operation. By leveraging torque, WOB, vibration, pressure, and temperature data, operators could make informed adjustments in real time. The ability to dynamically modify penetration speed based on scale presence further enhanced efficiency, ensuring optimal operational performance. These refinements resulted in reduced fluid consumption, minimized environmental impact, and significant cost savings, setting a new benchmark for coiled tubing descaling operations.

Key Performance Indicators (KPIs) for Descaling Operation Efficiency

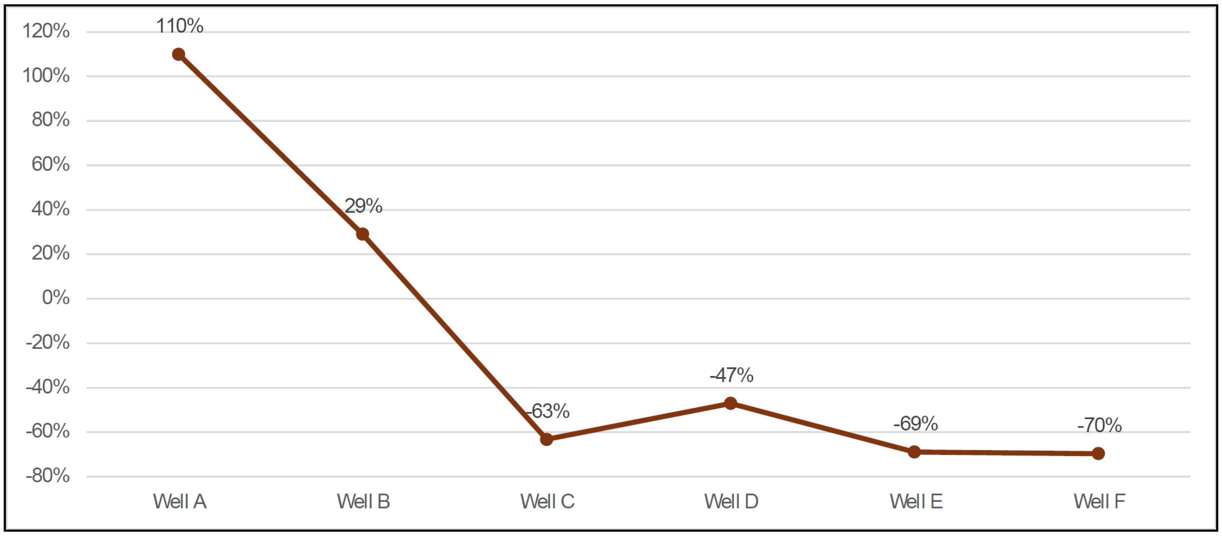
The optimization of descaling operations was measured against several key KPIs, including fluid consumption, water usage, nitrogen usage, Rate of Penetration (ROP), CT intervention time, and carbon emissions. The results from six wells (Well-A through Well-F) demonstrate significant improvements across all metrics, as summarized in the table and graph below.

Table 1—Descaling Operation Efficiency – Key Performance Indicators (KPIs) Across Six Wells

Well Number	Fluid Change	Change in Water Consumption	Change in Nitrogen Consumption	Change in ROP	Change in CT Time	Unnecessary Runs Eliminated	Change in Carbon Emission
Well-A	109.9%	117.5%	75%	–19%	19%	0	19%
Well-B	29%	29.7%	–70%	–30%	45%	0	45%
Well-C	–63.2%	–62.9%	–50%	68%	–52%	1 (Blasting)	–52%
Well-D	–47.1%	–46.2%	–58%	101%	–55%	1 (Blasting)	–55%
Well-E	–68.9%	–68.5%	–9%	91%	–49%	1 (Blasting)	–49%
Well-F	–69.7%	–69.7%	–63%	200%	–66%	1 (Blasting)	–66%

KPI: Reduction in Total Fluid Volume

Total fluid volume is a critical measure of CT operational efficiency. As shown in Graph 3, early wells (A and B) exceeded the designed fluid volumes due to trial-phase uncertainties. In contrast, Wells C through F demonstrated a marked reduction—up to 70%—reflecting improved control based on real-time telemetry.

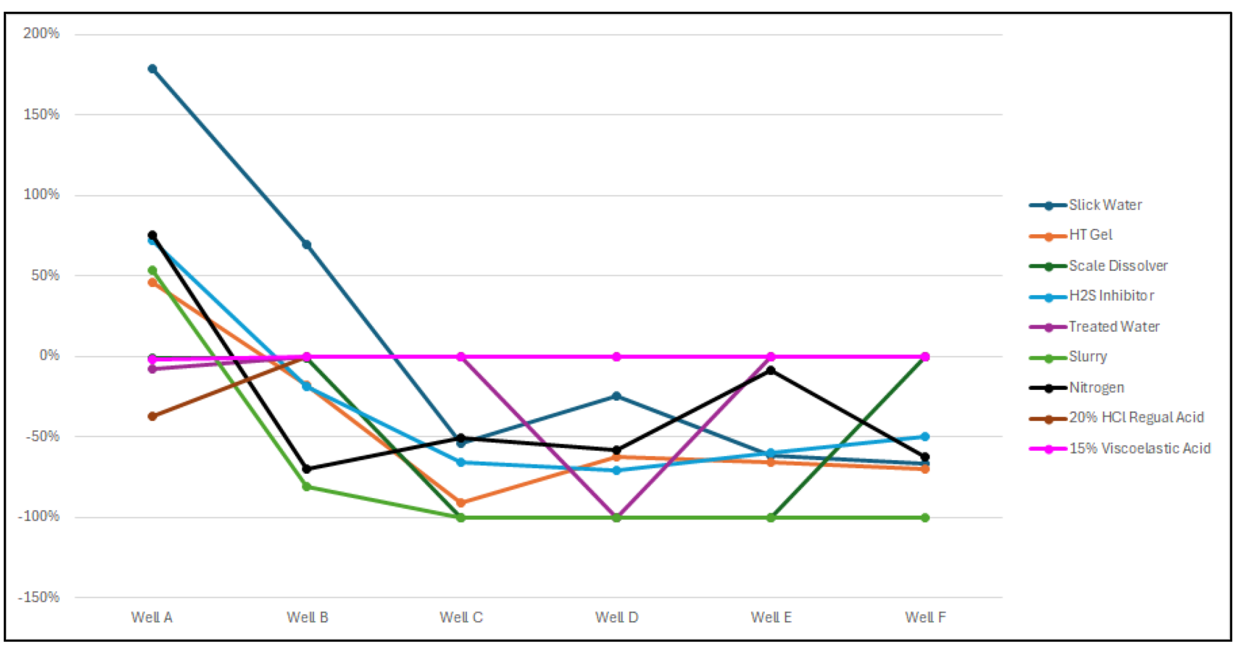


Graph 3—Total Fluid Volume Change after Utilizing Downhole Data

Significance: Lower fluid volumes reduce operational costs, logistical complexity, and environmental impact.

KPI: Change in Consumption per Fluid Type

Analyzing fluid usage by type rather than total volume provided better insights into job efficiency. In Graph 4, early operations used excessive slickwater and HT gel. Later wells showed more consistent and reduced consumption of scale dissolvers, indicating improved fluid planning based on downhole data.

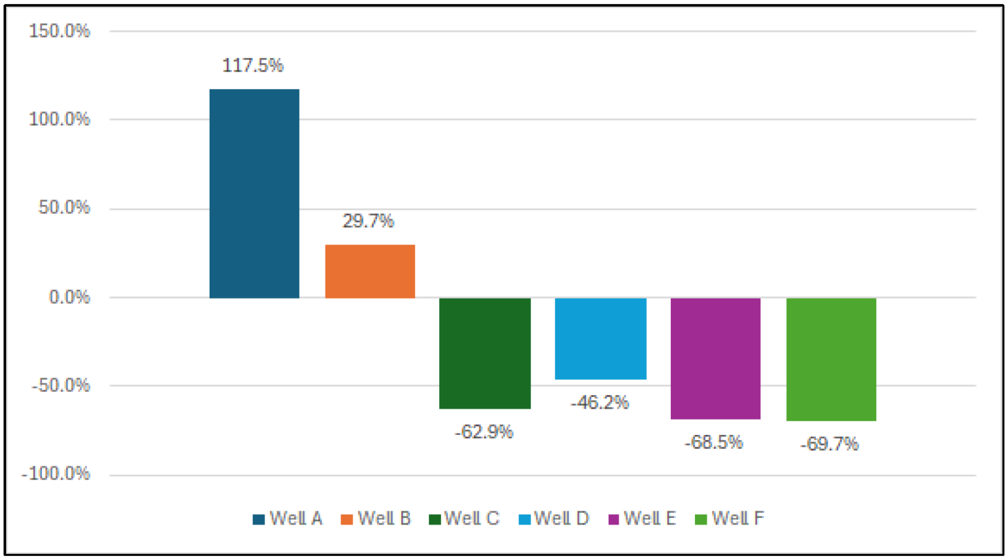


Graph 4—Fluid Volume Change Based on Fluid Type after Utilizing Downhole Data

Significance: Selective fluid use ensures efficient operations and minimizes chemical waste.

KPI: Change in Water Consumption

Water usage, shown in Graph 5, was significantly reduced in later wells—by up to 69.7%. Early wells exceeded design volumes, but subsequent improvements in pumping stages, rate control, and better scale removal strategies led to more sustainable water use.

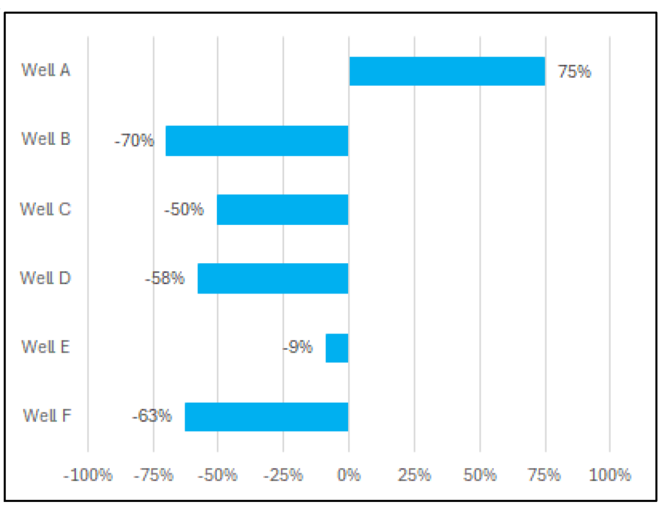


Graph 5—Water consumption changes compared to the initial design

Significance: Efficient water management supports environmental goals and resource conservation in water-scarce regions.

KPI: Change in Nitrogen Consumption

Nitrogen supply posed logistical challenges. As depicted in Graph 6, Well A required excessive nitrogen due to operational inefficiencies. In contrast, nitrogen usage in Wells C through F was reduced by up to 70% as telemetry enabled tighter pressure control and flow adjustments.

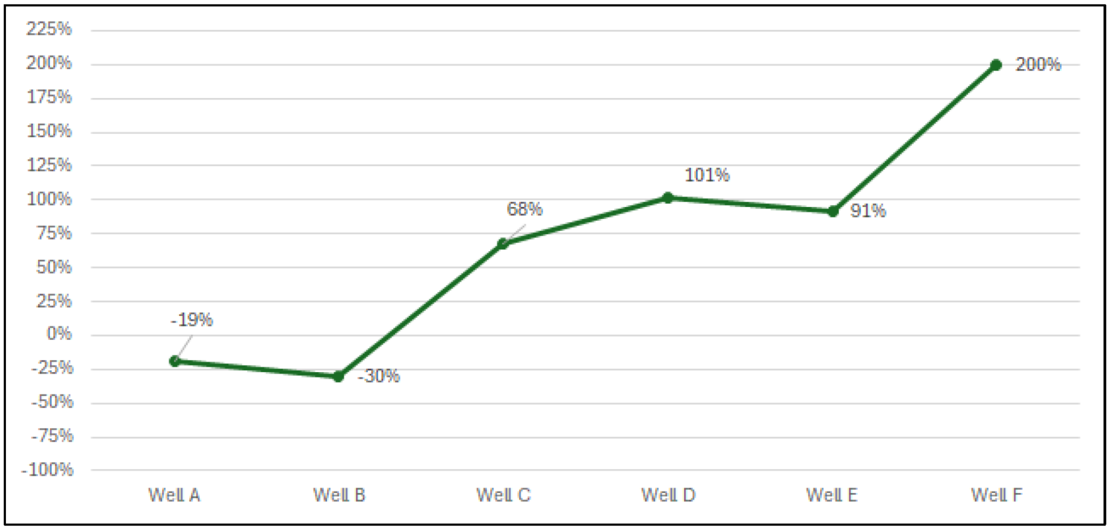


Graph 6—Nitrogen consumption changes compared to the initial design

Significance: Efficient nitrogen use lowers cost, minimizes downtime, and reduces site emissions.

KPI: Change in Rate of Penetration

ROP directly affects intervention duration. Graph 7 highlights up to a 200% improvement in Wells D and F due to optimized speed and torque control. Declines in Wells A and B reflect initial adoption challenges before full telemetry integration.

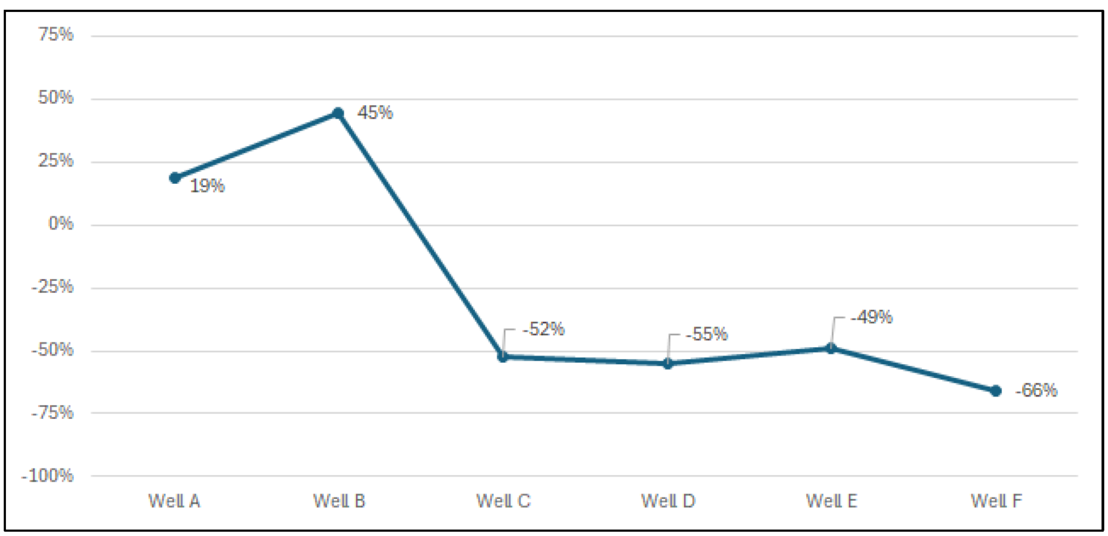


Graph 7—Rate of Penetration Change

Significance: Higher ROP improves job efficiency and accelerates well return to production.

KPI: Change in CT Intervention Time

A key operational metric, CT time in hole was reduced by up to 66% in later wells (see Graph 8). This was made possible by eliminating unnecessary trips, improving scale removal on the first pass, and increasing penetration speed where possible.



Graph 8—Change in CT Intervention Time

Significance: Shorter interventions reduce cost, equipment wear, and HSE exposure.

KPI: Change in Number of Necessary CT Runs

In most of the cases, the outside diameter (OD) of the bit used during descaling operations decreased. While this initially appeared as a limitation, it actually served as a key indicator that blasting with a slurry, a common technique for scale removal, could be avoided. This realization was particularly important when considering the potential for future well interventions, such as an E-line run to isolate lower completion for workover or maintenance activities. By observing the reduction in bit diameter and analyzing its implications, it became clear that the need for blasting could be postponed or, in some cases, completely eliminated as isolating the lower completion can be done with a plug with OD less than the OD of the bit after descaling.

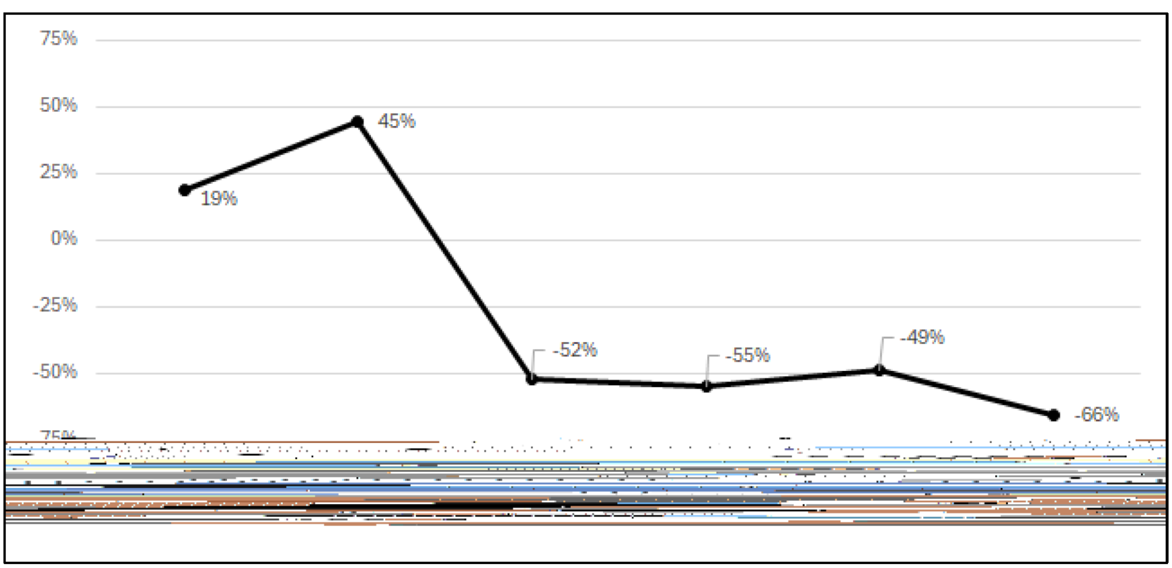
This shift in strategy not only minimized the immediate operational costs but also enhanced the long-term flexibility of the well. In future interventions, there would be no need to deal with the complications that often arise from prior blasting, such as debris accumulation or potential damage to the wellbore. This decision was made in recognition of the fact that a less invasive approach would maintain access to the lower completion and allow for more efficient and cost-effective future operations.

Furthermore, avoiding unnecessary blasting aligns with the broader goal of minimizing environmental impact. Blasting typically involves the release of particulate matter and other contaminants into the environment, which poses risks to surrounding ecosystems and requires extensive cleanup efforts. By postponing or eliminating this process, we were able to reduce the environmental footprint of the operation.

Ultimately, this proactive decision-making reflects a strategy focused not only on immediate operational efficiency but also on optimizing resource usage, reducing costs, and ensuring the well's long-term viability. By taking a more measured approach to scale removal and intervention planning, we were able to preserve the well's integrity, minimize unnecessary operations, and maintain the flexibility required for future interventions—ensuring that we are better prepared for any future challenges while maintaining a responsible and sustainable operational footprint.

KPI: Change in Carbon Emissions

As shown in Graph 9, carbon emissions dropped by up to 66% in optimized wells. These reductions were driven by lower diesel use due to fewer nitrogen delivery delays, reduced pumping durations, and fewer CT trips.



Graph 9—Change in carbon emissions

Significance: Lower emissions support regulatory compliance and demonstrate operational sustainability.

The combined reduction in water, nitrogen, and CO_2 emissions highlights the environmental advantages of a telemetry-driven approach. Real-time feedback enabled precise control over fluid and energy usage, helping operators meet sustainability targets without compromising performance

Discussion

The integration of real-time downhole telemetry into underbalanced descaling operations resulted in measurable improvements across operational, environmental, and reservoir-protection KPIs. Across six wells, telemetry enabled dynamic adjustments in real time, enhancing decision-making accuracy and minimizing inefficiencies.

Operationally, this approach yielded a 200% increase in rate of penetration and reduced coiled tubing intervention time by up to 66%. These gains were achieved through effective control of speed, torque, and fluid delivery based on live feedback. By eliminating unnecessary CT trips and improving first-pass effectiveness, overall job time and tool wear were substantially reduced.

From an environmental standpoint, the methodology reduced carbon emissions by as much as 66% and decreased water usage by 70% in optimized wells. These improvements align with global sustainability goals and reflect the efficiency of a data-informed approach to scale removal.

Maintaining underbalanced conditions throughout the operations was vital for reservoir protection. By keeping wellbore pressure below formation pressure, the system prevented dislodged scale from re-entering the reservoir, minimizing formation damage and preserving long-term productivity. This outcome is particularly critical in sour and low-pressure wells where formation integrity is at high risk.

While the learning curve was evident in initial wells, subsequent operations saw continuous refinement. Challenges such as nitrogen logistics were addressed through better supply planning and rate optimization. These improvements highlight the importance of integrating real-time telemetry not only into downhole systems but also into overall field execution strategy.

Looking forward, the methodology provides a strong foundation for automation and predictive analytics. Machine learning models could be trained on telemetry patterns—such as torque fluctuations, WOB variations, and vibration levels—to forecast optimal operational windows and detect early signs of inefficiency or tool failure. Such advancements would further reduce human intervention and enhance consistency across fields.

In summary, the successful deployment of real-time telemetry in underbalanced descaling operations represents a shift from reactive to proactive well intervention. It enables faster decision-making, safeguards the reservoir, reduces costs, and supports environmental commitments—offering a scalable and sustainable framework for future well interventions.

Conclusion

This study demonstrates how integrating real-time downhole telemetry into underbalanced coiled tubing descaling operations can significantly enhance performance, efficiency, and sustainability, particularly in sour and low-pressure environments. By enabling continuous monitoring of torque, WOB, vibration, pressure, and temperature, the approach transformed descaling from a trial-and-error method into a data-informed process. Operators could dynamically adjust operational parameters based on real-time insights, leading to tangible improvements across key performance indicators.

Across six field cases, this telemetry-based methodology delivered a 200% increase in rate of penetration, a 66% reduction in CT intervention time, and elimination of unnecessary runs. These gains were supported by improved fluid and energy management, reducing water usage by up to 70% and cutting nitrogen consumption by as much as 75%. As a result, total carbon emissions dropped by 66%, aligning with broader environmental targets without compromising operational outcomes.

Maintaining underbalanced conditions throughout the job played a central role in protecting reservoir integrity. Real-time pressure and temperature monitoring helped operators identify fluid losses early, adjust flow rates accordingly, and ensure efficient scale transport to surface. This minimized the risk of scale re-entry into the formation—a common issue with overbalanced techniques—and preserved long-term well performance.

In addition to immediate performance gains, the integration of real-time telemetry facilitated faster learning curves across operations. Teams became more efficient in interpreting data and making timely decisions, enhancing consistency and reducing variability between wells. Furthermore, by achieving cleanouts using smaller bit diameters without the need for blasting, the method also preserved future access to the lower completion, offering flexibility for future well interventions.

Looking ahead, this approach provides a strong foundation for the integration of machine learning and automation. Predictive models can be developed to detect scale buildup, optimize speed, and adjust fluid systems in real time, further reducing reliance on manual adjustments. The principles applied here can also be extended to other intervention challenges such as sand cleanouts or hydrate removal, expanding the impact of this methodology beyond scale management.

In summary, this work sets a new benchmark for coiled tubing operations in challenging well conditions. The telemetry-enabled underbalanced descaling strategy proves that high operational efficiency, environmental responsibility, and reservoir protection can be achieved simultaneously—offering a scalable solution for modern intervention challenges in mature fields.

Acknowledgment

The Authors would like to thank all TAQA's Coiled Tubing Operation team in making such an achievement possible with their dedication and resilience.

Nomenclature

CT = Coiled Tubing
BHA = Bottomhole Assembly
HSE = Health, Safety, & Environment
KPIs = Key Performance Indicators
ROP = Rate of Penetration
WOB = Weight on Bit

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Appendix

Carbon Emission Calculation for Coiled Tubing Operations

To estimate the carbon emissions generated by all the diesel-powered equipment in a typical coiled tubing operation, we will consider the fuel consumption of each piece of equipment and calculate the total emissions. Here is how we can proceed:

Diesel-Powered Equipment and Assumed Fuel Consumption:

1. Coiled Tubing Power Pack:
   Fuel consumption: ~12 gallons/hour
2. Generators (3 units):
   Fuel consumption: ~5 gallons/hour each
   Total: 3×5 = 15 gallons/hour
3. Nitrogen Pump:
   Fuel consumption: ~8 gallons/hour
4. Fluid Pump:
   Fuel consumption: ~6 gallons/hour
5. Air Compressor:
   Fuel consumption: ~4 gallons/hour
6. Other Diesel Equipment (e.g., crane, blender, auxiliary units):

Fuel consumption: ~6 gallons/hour (combined estimate)

Total Diesel Consumption:

12 (Power Pack) + 15 (Generators) + 8 (Nitrogen Pump) + 6 (Fluid Pump) + 4 (Air Compressor) + 6 (Other Equipment) = 51 gallons/hour

CO_2 Emissions per Gallon of Diesel:

Burning 1 gallon of diesel emits approximately 22.4 lbs (10.16 kg) of CO_2.

Total CO_2 Emissions:

51 gallons/hour × 22.4 lbs CO_2/gallon = 1,142.4 lbs CO_2/hour

Converting to metric units:

1,142.4 lbs × 0.4536 = 518.1 kg CO_2/hour

Final Estimate:

The operation generates approximately 518 kg (1,142 lbs) of CO_2 per hour, assuming the mentioned equipment and typical fuel consumption rates.